# 7.3 Permutations and Sign

An alternating form changes sign when two of its arguments are swapped (Theorem 7.2.1). To compare $f(v_3, v_1, v_2)$ with $f(v_1, v_2, v_3)$, we reach one list from the other by swaps and collect a factor $-1$ per swap. But the same rearrangement can be reached by different sequences of swaps, and Chapter 0 showed that even their number is not unique. If one route used an even number of swaps and another an odd number, we would get $f(v_1, v_2, v_3) = -f(v_1, v_2, v_3)$ for every alternating form, which is absurd for signed volume. This section rules that out. It attaches to each permutation a sign defined without any choices, proves that the sign is multiplicative, and deduces that the parity of the number of transpositions is well defined, as Chapter 0 promised.

## 7.3.1 Permutations, recalled

Recall from Chapter 0 that $S_n$ is the group of bijections $\{1, \dots, n\} \to \{1, \dots, n\}$, with $n!$ elements. A permutation can be written in two-line notation, with $\sigma(i)$ below $i$, or as a product of cycles. The product $\sigma\tau$ is the composition $\sigma \circ \tau$: **first** $\tau$**, then** $\sigma$, so products of cycles are read from right to left (Example (Computing with cycles)). A transposition $(i\ j)$ swaps $i$ and $j$ and fixes everything else, and for $n \geq 2$ every permutation is a product of transpositions (Theorem 0.10.4). The product is far from unique, since

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3) = (1\ 3)(1\ 2)(1\ 2)(1\ 2).$$

Here the numbers of transpositions are $2, 2, 4$, all even. The goal is to show this is no accident.

Why can we not simply **define** the sign of $\sigma$ as $(-1)^r$, where $\sigma$ is a product of $r$ transpositions? Because that definition depends on a choice of product, and we would have to prove that every choice gives the same answer. That is the statement we are trying to prove. We need a definition that involves no choice at all.

## 7.3.2 Inversions and the sign

To measure how much a permutation scrambles $1, 2, \dots, n$, look at the bottom row $\sigma(1), \dots, \sigma(n)$ of its two-line notation and count the pairs of entries that are in the wrong order. A sorted row has none, and each swap of two neighboring entries changes the count by exactly one. So the count is a natural candidate for tracking the parity of swaps.

*The sign of a permutation records whether it puts an even or an odd number of pairs out of order.*

**Definition 7.3.1. Inversion**

Let $\sigma \in S_n$. An **inversion** of $\sigma$ is a pair $(i, j)$ of positions with $1 \leq i < j \leq n$ **and** $\sigma(i) > \sigma(j)$. We write $\mathrm{inv}(\sigma)$ for the number of inversions of $\sigma$.

**Definition 7.3.2. Sign of a Permutation**

The **sign** of $\sigma \in S_n$ is

$$\operatorname{sgn}\sigma = (-1)^{\mathrm{inv}(\sigma)} \in \{1, -1\}.$$

The permutation $\sigma$ is **even** if $\operatorname{sgn}\sigma = 1$ and **odd** if $\operatorname{sgn}\sigma = -1$.

In words: an inversion is a pair of positions whose order $\sigma$ **reverses**. The condition $i < j$ makes

each pair of positions count once, and $\sigma(i) > \sigma(j)$ says the values come out in the opposite order. Since $\mathrm{inv}(\sigma)$ is the size of a subset of the finite set of pairs $i < j$, it is a well-defined integer between 0 and $n(n-1)/2$, and there is nothing to choose. That is the whole reason for this definition.

**The identity.** Its bottom row is sorted, so $\mathrm{inv}(\mathrm{id}) = 0$ and id is even. In $S_1 = \{\mathrm{id}\}$ this is the only permutation, with sign 1. The degenerate case matters: the identity is the empty product of transpositions, and its sign $1 = (-1)^0$ is what a product of 0 swaps should give.

**All of $S_3$.** Here is each permutation with its bottom row and inversions.

| $\sigma$ | id | (1 2) | (2 3) | (1 3) | (1 2 3) | (1 3 2) |
|---|---|---|---|---|---|---|
| bottom row | 1 2 3 | 2 1 3 | 1 3 2 | 3 2 1 | 2 3 1 | 3 1 2 |
| inversions | none | $(1,2)$ | $(2,3)$ | $(1,2)$, $(1,3)$, $(2,3)$ | $(1,3)$, $(2,3)$ | $(1,2)$, $(1,3)$ |
| sgn | 1 | $-1$ | $-1$ | $-1$ | 1 | 1 |

For (1 2 3), the bottom row is $\sigma(1), \sigma(2), \sigma(3) = 2, 3, 1$, and the out-of-order pairs of positions are $(1,3)$, since $2 > 1$, and $(2,3)$, since $3 > 1$.

Two things stand out. The three transpositions are odd, although (1 3) has three inversions, not one. And the two 3-cycles are even, consistent with (1 2 3) = (1 3)(1 2) being a product of two transpositions.

**Why all pairs, not just neighbors.** A tempting shortcut counts only **adjacent** positions $(i, i+1)$ with $\sigma(i) > \sigma(i+1)$. For (1 3), with bottom row $3, 2, 1$, this count is 2, which would make (1 3) "even". But (1 3) = (1 2)(2 3)(1 2), as one checks by following $1 \mapsto 2 \mapsto 3 \mapsto 3$, $2 \mapsto 1 \mapsto 1 \mapsto 2$, $3 \mapsto 3 \mapsto 2 \mapsto 1$, and each factor has adjacent count 1. A count that is to be multiplicative must give $(-1)^3 = -1$ here, so the adjacent count is the wrong one.

*Quick check.* Find sgn(1 4) in $S_4$ by counting inversions.

*Solution.* The bottom row is $4, 2, 3, 1$. The pairs of positions in the wrong order are $(1,2), (1,3), (1,4)$, since 4 is larger than everything after it, and $(2,4), (3,4)$, since $2 > 1$ and $3 > 1$. The pair $(2,3)$ is in order. So $\mathrm{inv}(1\ 4) = 5$ and $\mathrm{sgn}(1\ 4) = -1$.

## 7.3.3 The sign is multiplicative

The key property is that signs multiply when permutations are composed.

**Theorem 7.3.1. The Sign Is Multiplicative**

For all $\sigma, \tau \in S_n$,

$$\mathrm{sgn}(\sigma\tau) = \mathrm{sgn}\,\sigma \cdot \mathrm{sgn}\,\tau.$$

*Idea.* Count inversions one pair at a time, and record each pair as $+1$ (order kept) or $-1$ (order reversed). The sign is the product of these $\pm 1$'s. Applying $\tau$ and then $\sigma$ moves a pair of positions $\{i, j\}$ to $\{\tau(i), \tau(j)\}$ and then to $\{\sigma\tau(i), \sigma\tau(j)\}$. The order is reversed overall exactly when it is reversed at one of the two stages but not both, which is multiplication of $\pm 1$'s. Finally, as $\{i, j\}$ runs over all pairs, so does $\{\tau(i), \tau(j)\}$, so the contributions of $\sigma$ multiply to $\operatorname{sgn}\sigma$, just in a different order.

*Proof.* For a nonzero integer $m$ let $s(m) = 1$ if $m > 0$ and $s(m) = -1$ if $m < 0$. Then $s(mm') = s(m)s(m')$ for nonzero integers $m, m'$, by the sign rules for products of integers.

Let $\mathcal{P}$ be the set of 2-element subsets $\{i, j\}$ of $\{1, \dots, n\}$. For $\rho \in S_n$ and $\{i, j\} \in \mathcal{P}$, define

$$\varepsilon_\rho(\{i, j\}) = s\big((\rho(i) - \rho(j))(i - j)\big).$$

This is defined: $i \neq j$ and $\rho(i) \neq \rho(j)$ since $\rho$ is injective, so the product is nonzero; and exchanging the names $i$ and $j$ negates both factors, so the product does not depend on the order in which the subset is written. If $i < j$, then $i - j < 0$, so $\varepsilon_\rho(\{i, j\}) = -1$ exactly when $\rho(i) > \rho(j)$, that is, when $(i, j)$ is an inversion of $\rho$. Hence, by Definition 7.3.2,

$$\operatorname{sgn}\rho = \prod_{P \in \mathcal{P}} \varepsilon_\rho(P). \tag{1}$$

Now let $\{i, j\} \in \mathcal{P}$. Since $\tau(i) \neq \tau(j)$, we have $(\tau(i) - \tau(j))^2 > 0$. Multiplying inside $s$ by this positive integer does not change the value, so

$$\begin{aligned} \varepsilon_{\sigma\tau}(\{i, j\}) &= s\Big(\big(\sigma(\tau(i)) - \sigma(\tau(j))\big)\big(\tau(i) - \tau(j)\big) \cdot \big(\tau(i) - \tau(j)\big)(i - j)\Big) \\ &= \varepsilon_\sigma(\{\tau(i), \tau(j)\})\ \varepsilon_\tau(\{i, j\}). \end{aligned}$$

Let $\hat{\tau}\colon \mathcal{P} \to \mathcal{P}$, $\{i, j\} \mapsto \{\tau(i), \tau(j)\}$; the image has two elements because $\tau$ is injective. The same construction with $\tau^{-1}$ gives an inverse function, so $\hat{\tau}$ is a bijection (Theorem 0.4.7). Taking the product over all $P \in \mathcal{P}$ and using (1) three times,

$$\operatorname{sgn}(\sigma\tau) = \prod_{P \in \mathcal{P}} \varepsilon_\sigma(\hat{\tau}(P)) \cdot \prod_{P \in \mathcal{P}} \varepsilon_\tau(P) = \prod_{Q \in \mathcal{P}} \varepsilon_\sigma(Q) \cdot \operatorname{sgn}\tau = \operatorname{sgn}\sigma \cdot \operatorname{sgn}\tau,$$

where the second equality substitutes $Q = \hat{\tau}(P)$, which runs over $\mathcal{P}$ exactly once because $\hat{\tau}$ is a bijection, and reorders the finite product of integers. This proves the theorem.

■

By induction on the number of factors, $\operatorname{sgn}(\sigma_1\sigma_2\cdots\sigma_r) = \operatorname{sgn}\sigma_1 \cdots \operatorname{sgn}\sigma_r$. In the language of Chapter 0, the set $\{1, -1\}$ is a group under multiplication (it contains 1, and it is closed under products and inverses, since $(-1)^{-1} = -1$), and the theorem says exactly that

$$\operatorname{sgn}\colon S_n \to \{1, -1\}$$

is a group homomorphism (Definition 0.10.3).

## 7.3.4 Transpositions and parity

Now we can compute the sign of the building blocks.

**Corollary 7.3.2. Sign of a Transposition**

Every transposition is odd: $\operatorname{sgn}(i\ j) = -1$ for all $i \neq j$ in $\{1, \ldots, n\}$.

*Proof.* Since $(i\ j) = (j\ i)$, we may assume $i < j$. Let $\tau = (i\ j)$ and let $p < q$ be positions. We decide when $\tau(p) > \tau(q)$.

- If neither $p$ nor $q$ lies in $\{i, j\}$, then $\tau(p) = p < q = \tau(q)$: no inversion.
- If $(p, q) = (i, j)$, then $\tau(i) = j > i = \tau(j)$: one inversion.
- If $p = i$ and $q \neq j$, then $\tau(p) = j$ and $\tau(q) = q$, an inversion exactly when $i < q < j$. This gives $j - i - 1$ inversions.
- If $q = j$ and $p \neq i$, then $\tau(p) = p$ and $\tau(q) = i$, an inversion exactly when $i < p < j$. This gives $j - i - 1$ inversions.
- If $p = j$ (so $q > j$), then $\tau(p) = i < q = \tau(q)$; if $q = i$ (so $p < i$), then $\tau(p) = p < j = \tau(q)$: no inversion.

These cases cover every pair, so $\operatorname{inv}(\tau) = 1 + 2(j - i - 1)$, an odd number. Hence $\operatorname{sgn} \tau = -1$.

■

The Quick check above is the case $(i, j) = (1, 4)$: $1 + 2 \cdot 2 = 5$ inversions. Combining the corollary with multiplicativity gives the fact that Chapter 0 left open.

**Corollary 7.3.3. Parity Is Well Defined**

Let $n \geq 2$ and $\sigma \in S_n$. If $\sigma = \tau_1 \cdots \tau_r$ is a product of $r$ transpositions, then $\operatorname{sgn} \sigma = (-1)^r$. Consequently, if also $\sigma = \tau'_1 \cdots \tau'_s$ with transpositions $\tau'_m$, then $r$ and $s$ are both even or both odd.

*Proof.* By Theorem 7.3.1, extended to $r$ factors, and Corollary 7.3.2, $\operatorname{sgn} \sigma = \operatorname{sgn} \tau_1 \cdots \operatorname{sgn} \tau_r = (-1)^r$. In the same way $\operatorname{sgn} \sigma = (-1)^s$. Hence $(-1)^r = (-1)^s$, so $r - s$ is even.

■

So a permutation is even exactly when it is a product of an even number of transpositions, and the "number of swaps" has a well-defined parity even though the number itself is not defined.

**Proposition 7.3.4. Sign of a Cycle**

A $k$-cycle has sign $(-1)^{k-1}$. In particular, cycles of even length are odd and cycles of odd length are even.

*Proof.* For $k = 1$ the cycle is id, with sign $1 = (-1)^0$. Let $k \geq 2$ and let $\gamma = (i_1\ i_2\ \cdots\ i_k)$. We claim

$$\gamma = (i_1\ i_k)(i_1\ i_{k-1}) \cdots (i_1\ i_3)(i_1\ i_2).$$

Apply the factors from right to left. The element $i_1$ is sent to $i_2$ by $(i_1\ i_2)$, and no later factor involves $i_2$. For $2 \leq m \leq k - 1$, the element $i_m$ is fixed by the factors before $(i_1\ i_m)$,

sent to $i_1$ by it, then to $i_{m+1}$ by $(i_1\ i_{m+1})$, and no later factor involves $i_{m+1}$. The element $i_k$ is fixed until the last factor $(i_1\ i_k)$ sends it to $i_1$. Elements outside the cycle are fixed by every factor. This agrees with $\gamma$ everywhere, proving the claim. The product has $k-1$ transpositions, so $\operatorname{sgn}\gamma = (-1)^{k-1}$ by Corollary 7.3.3.

■

*Warning.* **Only the parity of the number of transpositions is determined, and it is not the parity of the cycle length.** The identity in $S_3$ equals $(1\ 2)(1\ 2)$ and also $(1\ 2)(1\ 3)(1\ 2)(1\ 3)(1\ 2)(1\ 3)$, since $((1\ 2)(1\ 3))^3 = (1\ 3\ 2)^3 = \mathrm{id}$; the numbers 2 and 6 differ, only their parity agrees. And a 3-cycle, "of length 3", is **even**: $(1\ 2\ 3) = (1\ 3)(1\ 2)$. One more trap concerns order: $(1\ 3)(1\ 2) = (1\ 2\ 3)$ but $(1\ 2)(1\ 3) = (1\ 3\ 2)$. Multiplying in the wrong order gives the wrong permutation and the wrong bottom row, even though, by Theorem 7.3.1, it happens to give the right sign.

## 7.3.5 The alternating group

Since sgn is a homomorphism, its kernel is a subgroup of $S_n$ (Theorem 0.10.3).

**Definition 7.3.3. Alternating Group**

The **alternating group** $A_n$ is the kernel of $\operatorname{sgn}\colon S_n \to \{1,-1\}$, that is, the subgroup of **even** permutations in $S_n$.

For example, $A_3 = \{\mathrm{id}, (1\ 2\ 3), (1\ 3\ 2)\}$ by the table above, and $A_2 = \{\mathrm{id}\}$. For $n \geq 2$, exactly half of the permutations are even (Exercise C1). The odd permutations do **not** form a subgroup: they do not contain id. The name is no coincidence: as we see below and in the next section, rearranging the arguments of an alternating form by an even permutation leaves its value unchanged.

## 7.3.6 Back to alternating forms

Here is why this section exists. Let $f$ be an alternating 3-linear form and compare $f(v_3, v_1, v_2)$ with $f(v_1, v_2, v_3)$. Swapping the first two arguments of $(v_3, v_1, v_2)$ gives $(v_1, v_3, v_2)$, and swapping the last two then gives $(v_1, v_2, v_3)$. By Theorem 7.2.1 (a) each swap costs a factor $-1$, so $f(v_3, v_1, v_2) = (-1)^2 f(v_1, v_2, v_3) = f(v_1, v_2, v_3)$. The list $(v_3, v_1, v_2)$ is $(v_{\sigma(1)}, v_{\sigma(2)}, v_{\sigma(3)})$ for $\sigma = (1\ 3\ 2)$, with bottom row $3, 1, 2$, and the factor 1 we found is $\operatorname{sgn}\sigma$.

A different sequence of swaps could have been used, possibly a longer one. By Corollary 7.3.3, every such sequence has an even number of swaps, so every route gives the same factor. In general, the next section proves that $f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = \operatorname{sgn}\sigma \cdot f(v_1, \ldots, v_k)$ for every alternating $k$-linear form, and uses it to turn each of the $n!$ surviving terms of the expansion sketched in the first section of this chapter into $\pm D(\mathbf{e}_1, \ldots, \mathbf{e}_n)$. Without the corollary, the factor computed swap by swap could depend on the route, and could not be identified with $\operatorname{sgn}\sigma$.

## 7.3.7 Computing signs in practice

There are now two independent ways to find a sign: count inversions, or factor into cycles and use Proposition 7.3.4 with multiplicativity. Doing both is a good check.

**Example. A Sign Computed Two Ways**

Find the sign of the permutation $\sigma \in S_6$ with two-line notation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 5 & 3 & 6 & 2 & 1 \end{pmatrix}$, first by counting inversions and then from its cycles.

*Solution. Inversions.* Go through the bottom row $4, 5, 3, 6, 2, 1$ from left to right, and for each entry count the **later** entries that are smaller:

| position $i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\sigma(i)$ | 4 | 5 | 3 | 6 | 2 | 1 |
| later and smaller | 3, 2, 1 | 3, 2, 1 | 2, 1 | 2, 1 | 1 | none |
| count | 3 | 3 | 2 | 2 | 1 | 0 |

So $\operatorname{inv}(\sigma) = 3+3+2+2+1+0 = 11$, which is odd, and $\operatorname{sgn}\sigma = -1$.

*Cycles.* Following elements as in Example (Computing with cycles): $1 \mapsto 4 \mapsto 6 \mapsto 1$, then $2 \mapsto 5 \mapsto 2$, and $3 \mapsto 3$. So $\sigma = (1\ 4\ 6)(2\ 5)$. By Proposition 7.3.4, the 3-cycle has sign 1 and the transposition has sign $-1$, so by Theorem 7.3.1, $\operatorname{sgn}\sigma = 1 \cdot (-1) = -1$. The two methods agree.

The following cell checks the example, and also tests Theorem 7.3.1 and Proposition 7.3.4 on all of $S_5$ by brute force. Permutations are stored as tuples of values $(\sigma(1), \ldots, \sigma(n))$, shifted to start at 0.

```
from itertools import permutations, combinations

def inv(s):
    return sum(1 for i, j in combinations(range(len(s)), 2) if s[i] > s[j])

def sgn(s):
    return (-1) ** inv(s)

def compose(s, t):          # (s t)(i) = s(t(i)): first t, then s
    return tuple(s[t[i]] for i in range(len(t)))

def cycle(entries, n):      # the cycle (entries[0] entries[1] ...), 0-based
    s = list(range(n))
    for a, b in zip(entries, entries[1:] + entries[:1]):
        s[a] = b
    return tuple(s)

sigma = (4, 5, 3, 6, 2, 1)
sigma0 = tuple(x - 1 for x in sigma)
print("inversions:", inv(sigma0), " sign:", sgn(sigma0))
print("(1 4 6)(2 5) equals sigma:", compose(cycle([0, 3, 5], 6), cycle([1, 4], 6))
```

```
    == sigma0)

S5 = list(permutations(range(5)))
print("multiplicative on S_5:", all(sgn(compose(s, t)) == sgn(s) * sgn(t) for s in
    S5 for t in S5))
print("k-cycles have sign (-1)^(k-1):", all(sgn(cycle(list(range(k)), 5)) == (-1) **
     (k - 1) for k in range(1, 6)))
```

*Run this code online*

# 7.3.8 Exercises

### 7.3.8.1 A. Check your understanding

**Exercise. A1**

(a) Define an inversion of $\sigma \in S_n$ and the sign $\operatorname{sgn}\sigma$.

(b) True or false: every 3-cycle is odd. Justify your answer.

(c) True or false: some permutation is a product of 5 transpositions and also a product of 8 transpositions. Justify your answer.

(d) What is the kernel of $\operatorname{sgn}\colon S_n \to \{1,-1\}$? Is the set of odd permutations a subgroup?

(e) True or false: $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\tau\sigma)$ for all $\sigma, \tau \in S_n$, even when $\sigma\tau \neq \tau\sigma$. Justify your answer.

(f) Let $f$ be an alternating 3-linear form. Express $f(v_2, v_3, v_1)$ in terms of $f(v_1, v_2, v_3)$.

*Solution.*

(a) An inversion of $\sigma$ is a pair $(i, j)$ with $1 \le i < j \le n$ and $\sigma(i) > \sigma(j)$ (Definition 7.3.1); $\operatorname{sgn}\sigma = (-1)^{\operatorname{inv}(\sigma)}$ (Definition 7.3.2).

(b) False. By Proposition 7.3.4 a 3-cycle has sign $(-1)^2 = 1$, so it is even; for instance $(1\ 2\ 3) = (1\ 3)(1\ 2)$.

(c) False. By Corollary 7.3.3 the numbers of transpositions in two such products have the same parity, and 5 and 8 do not.

(d) The kernel is the alternating group $A_n$ of even permutations (Definition 7.3.3). The odd permutations do not form a subgroup: id is even, so it is not among them.

(e) True. By Theorem 7.3.1 both sides equal $\operatorname{sgn}\sigma \cdot \operatorname{sgn}\tau$, and multiplication of integers is commutative. For example $(1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$, but both are 3-cycles, with sign 1.

(f) Swapping the first two arguments of $(v_2, v_3, v_1)$ gives $(v_3, v_2, v_1)$, and swapping the first and third then gives $(v_1, v_2, v_3)$. Each swap costs a factor $-1$ by Theorem 7.2.1 (a), so $f(v_2, v_3, v_1) = f(v_1, v_2, v_3)$. This matches the sign of the rearrangement $\sigma$ with bottom row $2, 3, 1$, namely $\sigma = (1\ 2\ 3)$, which is even.

### 7.3.8.2 B. Practice

**Exercise. B1: Computing Signs**

Find the sign of each permutation, both by counting inversions and by using cycles.

(a) $\sigma \in S_4$ with two-line notation $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$.

(b) $(1\ 2)(3\ 4\ 5) \in S_5$.

(c) $(1\ 2\ 3)(2\ 3\ 4) \in S_4$.

*Solution.*

(a) *Inversions.* In the bottom row $2, 4, 1, 3$, the entry 2 is followed by the smaller entry 1, the entry 4 by 1 and 3, and the entry 1 by nothing smaller. So $\operatorname{inv}(\sigma) = 1 + 2 + 0 = 3$ and $\operatorname{sgn}\sigma = -1$. *Cycles.* $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$, so $\sigma = (1\ 2\ 4\ 3)$, a 4-cycle, with sign $(-1)^3 = -1$ by Proposition 7.3.4.

(b) *Cycles.* By Proposition 7.3.4 and Theorem 7.3.1, the sign is $(-1) \cdot (-1)^2 = -1$. *Inversions.* The permutation sends $1 \mapsto 2$, $2 \mapsto 1$, $3 \mapsto 4$, $4 \mapsto 5$, $5 \mapsto 3$, so the bottom row is $2, 1, 4, 5, 3$. The inversions are $(1, 2)$, $(3, 5)$ and $(4, 5)$, so there are 3, and the sign is $-1$.

(c) *Cycles.* Both factors are 3-cycles, so the sign is $1 \cdot 1 = 1$. *Inversions.* Apply $(2\ 3\ 4)$ first: $1 \mapsto 1 \mapsto 2$, $2 \mapsto 3 \mapsto 1$, $3 \mapsto 4 \mapsto 4$, $4 \mapsto 2 \mapsto 3$. The bottom row is $2, 1, 4, 3$, with inversions $(1, 2)$ and $(3, 4)$. So the sign is $(-1)^2 = 1$, and in fact the product is $(1\ 2)(3\ 4)$.

**Exercise. B2: Sign of the Inverse**

Prove that $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}\sigma$ for every $\sigma \in S_n$.

*Solution.* Let $\sigma \in S_n$. By Theorem 7.3.1, $\operatorname{sgn}\sigma \cdot \operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$, since id has no inversions. Since $\operatorname{sgn}\sigma \in \{1, -1\}$ and $(\pm 1)^{-1} = \pm 1$, multiplying both sides by $\operatorname{sgn}\sigma$ gives $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}\sigma$.

**Exercise. B3: The Reversal Permutation**

Let $\rho \in S_n$ be the reversal, $\rho(i) = n + 1 - i$. Show that $\operatorname{inv}(\rho) = n(n-1)/2$. Hence find the $n$ for which $\rho$ is even, and decide the sign of $\rho$ for $n = 6$ and $n = 9$.

*Solution.* For $1 \le i < j \le n$, $\rho(i) = n + 1 - i > n + 1 - j = \rho(j)$. So **every** pair $i < j$ is an inversion, and $\operatorname{inv}(\rho)$ is the number of 2-element subsets of $\{1, \dots, n\}$, which is $n(n-1)/2$. Hence $\operatorname{sgn}\rho = (-1)^{n(n-1)/2}$.

Write $n = 4q + t$ with $t \in \{0, 1, 2, 3\}$. Then $n(n-1)/2$ is $2q(4q-1)$ for $t = 0$, $2q(4q+1)$ for $t = 1$, $(4q+1)(2q+1)$ for $t = 2$, and $(2q+1)(4q+3)$ for $t = 3$. The first two are even and the last two are odd. So $\rho$ is even exactly when $n$ leaves remainder 0 or 1 on division by 4. For $n = 6$, $\operatorname{inv}(\rho) = 15$ and $\rho$ is odd; for $n = 9$, $\operatorname{inv}(\rho) = 36$ and $\rho$ is even.

#### 7.3.8.3 C. Going deeper

**Exercise. C1: Half the Permutations Are Even**

Let $n \geq 2$. Prove that exactly $n!/2$ permutations in $S_n$ are even.

*Hint: consider* $\sigma \mapsto (1\ 2)\sigma$.

*Solution.* Let $E$ and $O$ be the sets of even and odd permutations; they are disjoint and $E \cup O = S_n$, so $|E| + |O| = n!$. Let $\tau = (1\ 2)$, which exists since $n \geq 2$, and define $L\colon S_n \to S_n$, $L(\sigma) = \tau\sigma$. By Theorem 7.3.1 and Corollary 7.3.2, $\operatorname{sgn} L(\sigma) = -\operatorname{sgn}\sigma$, so $L$ maps $E$ into $O$ and $O$ into $E$. Since $\tau\tau = \mathrm{id}$, $L(L(\sigma)) = \sigma$, so $L$ restricted to $E$ is a bijection $E \to O$ with inverse $L$ restricted to $O$ (Theorem 0.4.7). Hence $|E| = |O| = n!/2$.

**Exercise. C2: Sign from the Number of Cycles**

Suppose $\sigma \in S_n$ is written as a product $\sigma = \gamma_1\gamma_2\cdots\gamma_c$ of **disjoint** cycles (no element appears in two of them) whose lengths $k_1, \ldots, k_c$ add up to $n$, where fixed points are included as cycles of length 1. (Such a product always exists: follow each element around its loop, as in Example (Computing with cycles).)

(a) Prove that $\operatorname{sgn}\sigma = (-1)^{n-c}$.

(b) Use (a) to find the sign of $\sigma = (1\ 4\ 6)(2\ 5) \in S_6$, and compare with Example (A Sign Computed Two Ways).

(c) Deduce that an $n$-cycle in $S_n$ and the identity in $S_n$ have the same sign exactly when $n$ is odd.

*Solution.*

(a) By Theorem 7.3.1 and Proposition 7.3.4,

$$\operatorname{sgn}\sigma = \prod_{m=1}^{c} \operatorname{sgn}\gamma_m = \prod_{m=1}^{c} (-1)^{k_m - 1} = (-1)^{(k_1+\cdots+k_c)-c} = (-1)^{n-c}.$$

(b) Including the fixed point 3 as the cycle $(3)$, $\sigma = (1\ 4\ 6)(2\ 5)(3)$, with lengths $3+2+1 = 6$ and $c = 3$. So $\operatorname{sgn}\sigma = (-1)^{6-3} = -1$, as in Example (A Sign Computed Two Ways).

(c) An $n$-cycle has $c = 1$ and sign $(-1)^{n-1}$; the identity consists of $n$ cycles of length 1, so $c = n$ and its sign is $(-1)^0 = 1$. These agree exactly when $n - 1$ is even, that is, when $n$ is odd.